

Experiment No:

Date: _____

Analyze the beamforming (Radiation Pattern) parameters of linear array antenna in wireless communication

Tools Required: MATLAB R2015a/ Simulink library

Theory: Beamforming is a signal processing technique that enhances the performance of wireless communication systems. It involves directing the transmission or reception of radio waves in a specific direction, rather than broadcasting signals in all directions equally. This targeted approach allows for improved signal quality, increased range, and reduced interference.

In beamforming, multiple antennas are used to transmit or receive signals. By manipulating the phase and amplitude of each antenna's signal, the system can create constructive interference in the desired direction and destructive interference in other directions. This results in a focused beam of energy that can be steered towards a specific user or location.

Beamforming technology has been widely adopted in various applications such as wireless communication systems (e.g., Wi-Fi routers), radar systems, and cellular networks. It offers several advantages including increased data rates, improved coverage, reduced power consumption, and enhanced overall system capacity.

Benefits of Beamforming

As 5G networks continue their global rollout, enabling smartphones, connected devices, and wide-area networking applications, beamforming emerges as a critical core technology. The high-frequency millimeter wave (mmWave) spectrum used in 5G communications is susceptible to disruptions from obstacles like walls and other barriers, posing challenges for reliable connectivity.

Beamforming plays a crucial role in mitigating these challenges by allowing transmitters to focus their signals in specific directions towards mobile devices, vehicles, or Internet of Things (IoT) devices. This directional transmission not only enhances signal strength but also improves the overall reliability of 5G connections.

Moreover, beamforming will be a key enabler for massive MIMO (Multiple-Input Multiple-Output) technology in 5G networks. Massive MIMO employs a large number of antennas at the base station, which, in conjunction with beamforming, can direct beams both horizontally and vertically towards user devices. This precise beam steering capability not only maximizes throughput but also optimizes spectral efficiency, leading to improved network performance and capacity.

```
clc;
clear;
close all;
```

```
% Parameters
frequency = 3e9; % Frequency (3 GHz)
lambda = 3e8 / frequency; % Wavelength
numElements = 8; % Number of elements in the array
d = lambda / 2; % Element spacing (in wavelengths)
theta_steering = 30; % Desired steering angle (degrees)
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% Angle range for radiation pattern
angles = linspace(-90, 90, 180); % Angles from -90 to 90 degrees

% Define the element positions in a linear array
elementPositions = (0:numElements-1) * d; % Positions of the
elements

% Calculate the array factor
AF = zeros(size(angles)); % Initialize array factor

for m = 1:numElements
    beta = exp(1j * (2 * pi / lambda) * (elementPositions(m) *
sind(theta_steering))); % Phase shift for steering
    AF = AF + beta * exp(1j * (2 * pi / lambda) *
(elementPositions(m) * sind(angles))); % Array factor
    calculation
end

% Normalize the array factor
AF = abs(AF);
AF = AF / max(AF); % Normalize to maximum value

% Plot the radiation pattern
figure;
polarplot(deg2rad(angles), AF, 'LineWidth',2);
title('Beamforming Pattern of Patch Antenna Array');
ax = gca;
ax.Theta = -90:30:90; % Set theta
grid on;

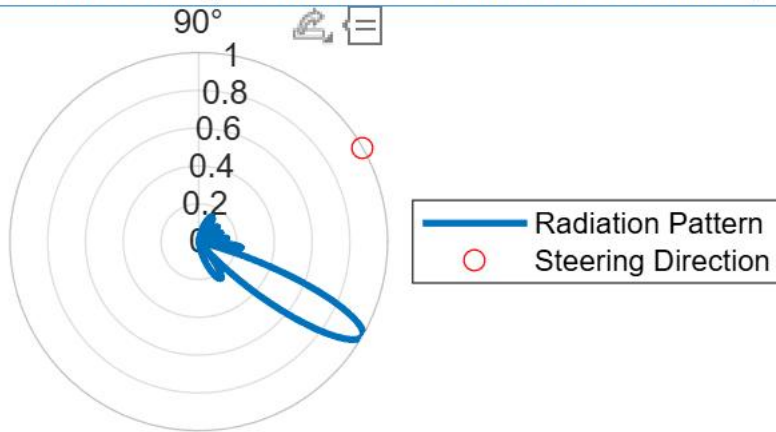
% Display the main lobe direction
hold on;
polarplot(deg2rad(theta_steering), max(AF), 'ro'); % Marking the
steering direction
legend('Radiation Pattern', 'Steering Direction');
hold off;

```

Observations:

Change the parameters **Desired steering angle** and observe the steering direction and main beam

Beamforming Pattern of Patch Antenna Array



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